

Feature Extraction and Signal Processing of IMU Data on a Raspberry Pi Pico

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Abstract—Three motions were recorded with an ICM-20948 at 500 Hz and processed on the Raspberry Pi Pico itself: a Butterworth low-pass, quantisation at 16, 8 and 4 bits, descriptive statistics and a 2048-point FFT. Gyroscope variance separates the motions by six orders of magnitude and the dominant spectral component identifies the shaking rate, while variance alone cannot distinguish walking from shaking. The on-device transform agrees with a NumPy reference to within 0.03 % of the largest bin, and the full pipeline consumes 3.9 % of the available sample-period budget.

Index Terms—IMU, feature extraction, quantisation, FFT, embedded systems

I. METHOD

The selected application domain is the inertial measurement unit. The Lab 1.1 logger samples the ICM-20948 over I²C at 500 Hz and writes 16-byte records to an SD card. Three recordings of 15 s were acquired: the board at rest, carried during walking, and shaken by hand, each of 7500 samples with none dropped.

A second program reads a recording back on the Pico, converts the raw counts to g and dps, and processes the first 2048 samples: DC removal and peak normalisation; a second-order Butterworth low-pass designed by bilinear transform and applied in the forward direction only; quantisation at 16, 8 and 4 bits with dequantisation and error measurement; per-axis mean, variance, standard deviation, minimum, maximum, median and mode; and a 2048-point radix-2 FFT with a Hamming window. Each stage is timed with `time_us_64()`. The analysed block and the resulting spectrum are written back as CSV, and a host-side script recomputes the transform in NumPy over identical samples. The Sobel and Canny operators were omitted as they are defined on images.

Two consistency checks validate the data. With the board flat the mean of a_z is -1.00048 g, i.e. gravity; the gyroscope means in the same recording, -0.195 , -0.106 and -0.142 dps, are the sensor bias.

II. RESULTS

Table I reports the variance of every axis across the three recordings; the accelerometer is expressed in g and the gyroscope in dps, so the two groups are not directly comparable in magnitude. Table II gives the effect of an identical 5 Hz cutoff on all three recordings, that value being chosen as an intermediate point between the walking band below 1.5 Hz

and the shaking component at 7.57 Hz. Table III gives the quantisation error for a_x of the shaking recording, the other two behaving equivalently. Table IV states the agreement between the on-device transform and the NumPy reference, with the deviation expressed relative to the largest bin of the reference spectrum, together with the measured cost of each processing stage. Fig. 1 presents all three recordings with their spectra and Fig. 2 overlays the two implementations for the shaking case.

TABLE I
VARIANCE PER AXIS, 2048 SAMPLES AT 500 HZ

Axis	still	walking	shaking
a_x [g ²]	0.0000	0.0087	0.0095
a_y [g ²]	0.0000	0.0070	0.0606
a_z [g ²]	0.0000	0.0143	0.0766
g_x [dps ²]	0.0013	127.7	3615.9
g_y [dps ²]	0.0010	129.9	1245.0
g_z [dps ²]	0.0010	1024.5	323.1

TABLE II
BUTTERWORTH LOW-PASS, $f_c = 5$ HZ, APPLIED TO a_x

Recording	Std before	Std after	Removed
still	0.000588	0.000427	27.3 %
walking	0.093340	0.091487	2.0 %
shaking	0.097659	0.056659	42.0 %

TABLE III
QUANTISATION OF a_x , SHAKING, 2048 SAMPLES

Bits	SNR [dB]	RMS error	Max error	Clipped
16	91.61	8.88×10^{-6}	1.53×10^{-5}	0
8	43.43	2.28×10^{-3}	3.91×10^{-3}	0
4	19.45	3.60×10^{-2}	6.24×10^{-2}	6

III. ANALYSIS

A. Feature selection

Gyroscope X variance spans 0.0013 at rest, 127.7 during walking and 3615.9 during shaking (Table I). The separation

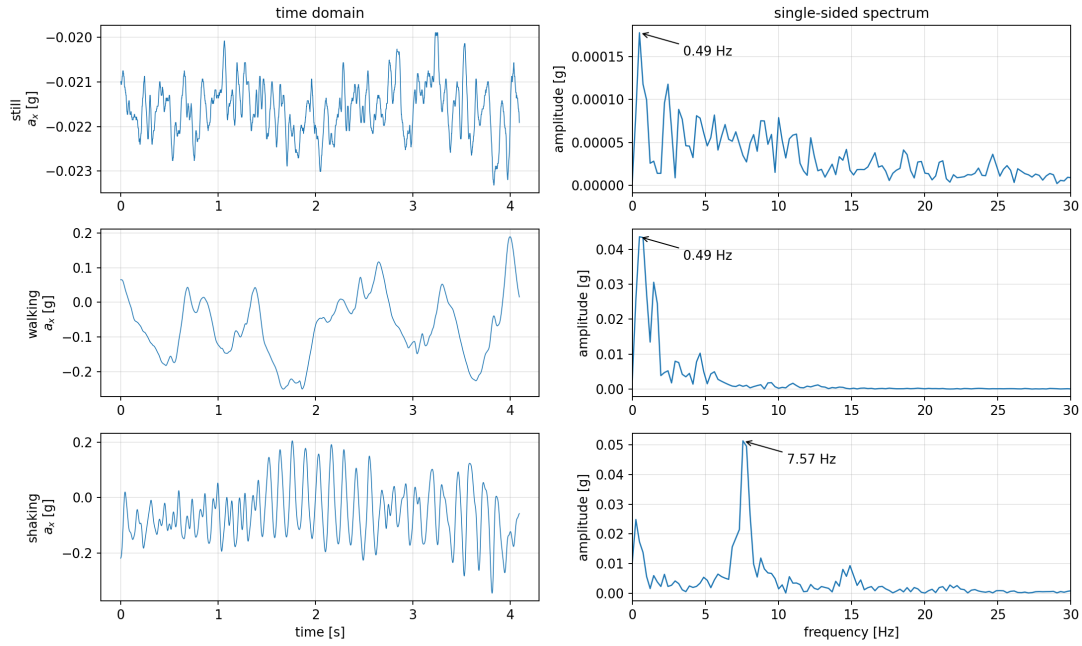


Fig. 1. Accelerometer X for the three recordings (left) and the corresponding single-sided spectra (right). The dominant bin is annotated.

TABLE IV
ON-DEVICE FFT AGAINST NUMPY, AND STAGE COST PER 2048-SAMPLE BLOCK

Recording	Peak [Hz]	Max diff.	Rel. diff.
still	0.49	4.1×10^{-8}	0.0231 %
walking	0.49	3.6×10^{-7}	0.0008 %
shaking	7.57	3.0×10^{-7}	0.0006 %
Low-pass		11.8 ms	5.8 μ s/sa.
Quantise + dequantise		12.0 ms	5.9 μ s/sa.
FFT, 2048 points		160.3 ms	78.3 μ s/sa.
Statistics, six axes		3.08 s	250 μ s/sa.

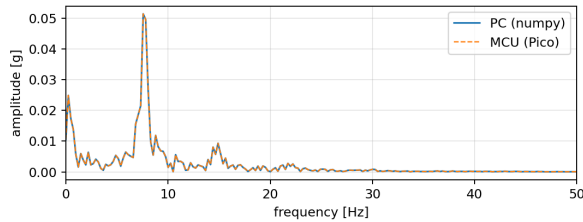


Fig. 2. Shaking recording. The on-device spectrum (dashed) coincides with the NumPy spectrum (solid) across the whole band.

approaches six orders of magnitude and exceeds the measurement noise by a wide margin, so any threshold between 1 and 50 distinguishes motion from rest without tuning.

Variance alone is nevertheless insufficient. The a_x variances for walking and shaking, 0.0087 and 0.0095, are effectively identical, whereas the spectra are not: walking concentrates its energy below 1.5 Hz, while shaking peaks at 7.57 Hz with a first harmonic near 15 Hz arising because a manual shake is

not sinusoidal (Fig. 1). Variance describes the magnitude of motion and the dominant spectral component its rate of repetition; the two are complementary and neither is individually sufficient.

The filter results in Table II are governed by the same consideration. Walking lies inside the 5 Hz passband and loses 2 % of its spread; the still recording contains only noise, of which 27 % is removed; shaking peaks above the cutoff and loses 42 %, taking the signal of interest with it. The cutoff is therefore a property of the motion under investigation, not of the sensor.

B. Quantisation and numerical accuracy

The SNR falls by 48.18 dB between 16 and 8 bits and by 23.98 dB between 8 and 4 bits, i.e. 6.02 and 6.00 dB per bit, reproducing the expected halving of quantisation error per additional bit.

The on-device FFT deviates from the NumPy reference by at most 3.6×10^{-7} , or 0.0008 % of the largest bin, with all five strongest peaks agreeing in bin index, frequency and amplitude to six decimal places (Table IV, Fig. 2). The residual is attributable to single- against double-precision arithmetic. The relative error is largest for the still recording only because the reference bin it is normalised against is itself very small.

C. Execution cost

The 2048-point transform requires 160.3 ms against the 4.096 s needed to acquire the same block, so processing occupies 3.9 % of the available time and a single core sustains both concurrently. The per-sample cost of 78.3 μ s places the theoretical ceiling near 12.8 kHz; in practice sensor readout and card writes reduce this, the worst observed `f_write`

being 3.9 ms against a 2 ms sample period, tolerable only because the logger buffers 4 kB per write and samples on the second core.

The statistics dominate at 3.08 s for six axes, almost entirely due to the median's $O(n^2)$ insertion sort. Mean, variance, minimum and maximum are single-pass and negligible. For a_x at rest the mean is -0.02161 and the median -0.02155 , a difference of 0.3 %, so the median does not justify its cost outside outlier rejection.

IV. ISSUES AND DISCUSSION

Clipping occurred in two distinct forms. The first shaking recording saturated the sensor: configured for ± 2 g, samples reached 32767, the 16-bit maximum. Such peaks are unrecoverable and the recording was repeated with a gentler motion; the appropriate remedy is a ± 8 or ± 16 g range, trading resolution for headroom. The second form is implementation-side. Normalisation places the largest sample at 0.999, which is representable at 16 bits ($32767/32768$) but not at 4 bits ($7/8 = 0.875$), where six samples fall outside. The normalisation target should be a function of bit depth; it was retained as the effect is instructive, though it inflates the 4-bit error slightly.

The mode was expected to be uninformative for continuous data on the assumption that measured values rarely repeat. The raw output is an integer count, and at rest the signal traverses so few counts that values repeat between 103 and 777 times; in the moving recordings this falls to between 3 and 7. The mode is thus informative only where the signal is nearly static, which is where it carries least information.

Two improvements follow. The median should be replaced by a counting approach over the integer counts, or dropped in favour of variance, eliminating almost all of the 3.08 s. Secondly, the analysis operates on a stored file rather than a live stream, which ensured repeatable results across experiments but does not demonstrate sustained real-time operation; executing the same code against the live sensor is the natural continuation.